Experimental Determination of Moment of Inertia

Alfred Petersson, Jonathan Johansson

Course
Mechanics
Uppsala University

May 16, 2026

Contents

1 Abstract

1.1

This report investigates the moment of inertia of rotating bodies using experimental measurements of angular velocity as a function of time. Linear regression was used to determine the angular acceleration from the measured data, and the results were compared for different experimental configurations. The aim was to examine how the distribution of mass affects the moment of inertia and to evaluate how well the experimental results agree with theoretical expectations.

2 Introduction

2.1

Moment of inertia is a physical property of anything which rotates, charachterized by the resistance to changes in angular velocity. Since this affects how much force needs to be applied to accelerate or decelerate objects, knowing how to calculate anc measure it is crucial for many applications. In this study several means of measuring the moment of inertia practically were explored and compared with theoretical values. The aim was to determine if this was a valid means of mesuring the moment of inertia and how accurate of a method it was. A second goal of the experiment was to determine the effect of impulse on a rotating system to show with a practical example that angular momentum is conserved in systems like this while angular kinetic energy is not.

3 Theory/Background

3.1

To calculate with rotational movement and can not idealize the mass to a particle as often done in translational calculations, you have to use expresions giving a link between the mass of an object and how it is distributed [2, p. 339]

$I = moment\,of\,inertia$, different geometric forms will give diferent calculations about its centroid being the center of the distributed mass.

Examples of equations of moment of inertia

Equation for a circular plate

$$I_x = I_y = \frac{1}{2}I_z = m\frac{r^2}{4} \tag{1}$$

[1, p. 185]

Equation for a circular ring

$$I_x = I_y = \frac{1}{4}m(r_1^2 + r_2^2) \tag{2}$$

[1, p. 185]

Using this link Newtons second law $\sum F = ma$ analogusly can be aplied for rotational movement [2, p. 335].

$$\sum \tau = I\alpha \tag{3}$$

τ equals torsion I equals moment of inertia and α the angular acceleration

To calculate Torque (τ) it is the distance to where the aplied Force perpendikular to the rotational center.

$$\tau = Tr \tag{4}$$

[1, p. 171]

T is the aplied force in Newton and r is the distance to the point of perpendicular aplied force.

Given an expresion for $\tau = Tr$ the two formulas can be joined to

$$Tr = I\alpha \tag{5}$$

In the case that the force is generated with a free faling weight the folowing substitutions can be made $\sum F = mg - T = ma$. a = the linear acceleration of the falling mass giving the expression.

$$mgr = I\alpha \tag{6}$$

When the centroid of an object and the axes of rotation do not aligne The Paralell axel Theorem (Steiners's Theorem) can be used to calculate the moment of inertia.

$$I_{total} = I_{cm} + mr^2 \; (r = the\, distance\, between\, axis) \tag{7}$$

[1, p. 181]

With the given equations a expression for the experimantal moment of inertia can be derived, taken in account that the moment of inertia are linear and can be summed the expression for the parallel theorem can be added.

$$I = \frac{mgr}{\alpha} + mr^2 \tag{8}$$

For calculations of angular mommentum L and rotational energy k_{rot} fomulas analogus to straight movement are used.

$$E_{rot} = \frac{1}{2}\omega I \tag{9}$$

$$L = I\omega \tag{10}$$

[1, p. 181]

For error estimation the standard deviation can be calculated to give a sense of how close to the mean value the mesurements are distributed. x_i is the experimental value, x the mean value and n the amount of data collections

$$S = \sqrt{\frac{\sum_{i=1}^{n}(x_i - x)^2}{n-1}} \tag{11}$$

[1, p. 463]

4 Method

4.1

Two differrent experiments were conducted one to experimentaly decide the moment of inertia and one to see what hapens with angular momenton and rotational energy after an impact with an aditional bod

5 Results

5.1

Kolumn 1	Kolumn 2	Kolumn 3
1	2	3
4	5	6

Table 1: Min tabell

fisk

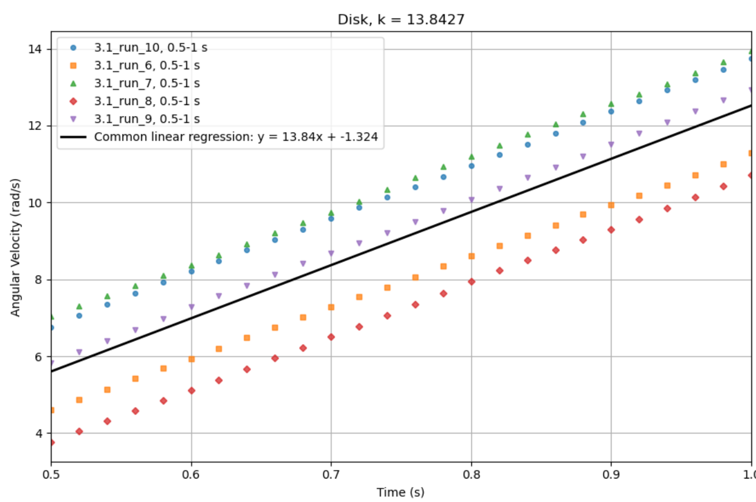

Figure 1: Graph from the experiment used to determine the moment of inertia of a disk. The k-value gives the angular acceleration.

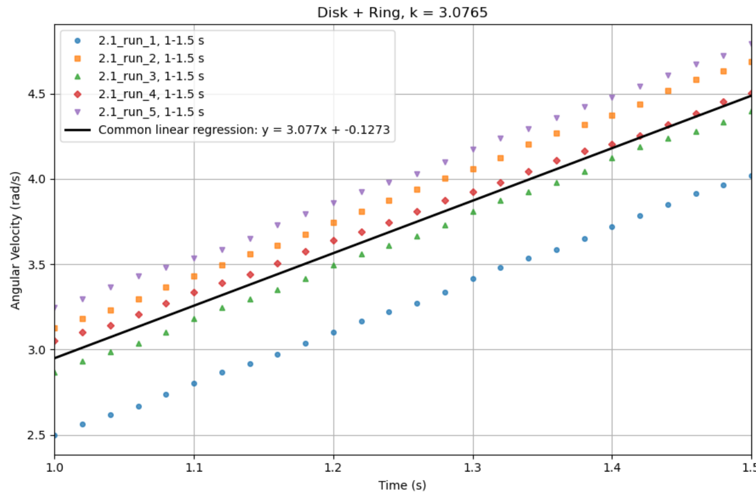

Figure 2: Graph from the experiment used to determine the moment of inertia of a disk with added weight. The k-value gives the angular acceleration.

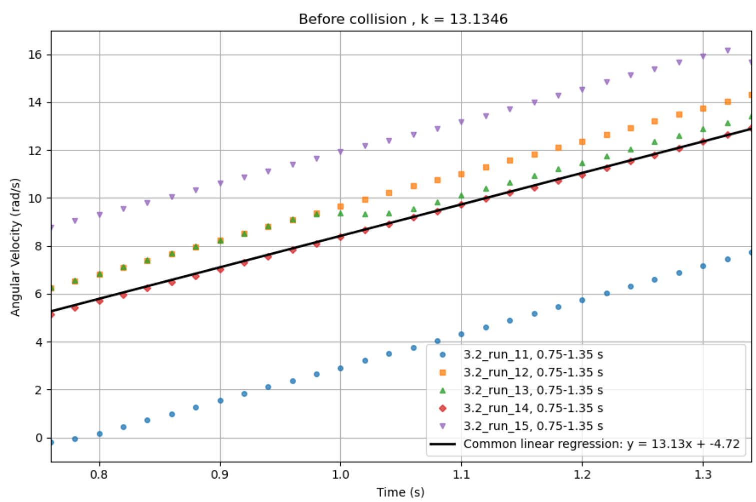

Figure 3: Graph showing the angular speed before impact of the added weight. The k-value gives the angular acceleration.

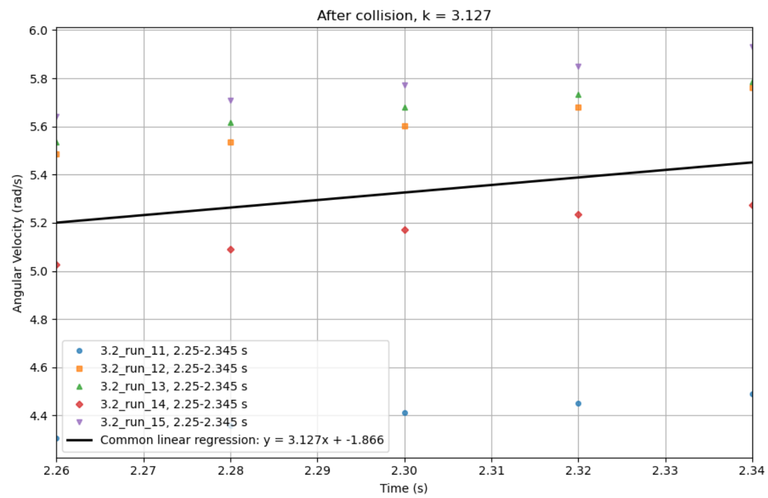

Figure 4: Graph showing the angular speed after impact of the added weight. The k-value gives the angular acceleration.

haj

6 Discussion

6.1

7 References

7.1

References

[1] Carl Nordling and Jonny Österman. *Physics Handbook for Science and Engineering*. 9th ed. Studentlitteratur, 2020.

[2] Hugh D. Young and Roger A. Freedman. *University Physics*. 11th ed. Pearson, 2003.

8 Appendix

8.1